

Clinical Utility (AUC-ROC, Higher is Better)

Demographic Parity Difference (DPD, Lower is Fairer)

Hierarchical MoE v2 (K=2)
Restores AUC (0.8261) & Net Benefit
(AUC Recovery +0.1238 vs v1)

Hard Cluster v1 (K=2)
Apparent Parity (DPD=0.0324)
Severe Data Fragmentation (AUC=0.7022)

- Single Model (Logistic K=1)
- Single Model (LightGBM K=1)
- Single Model (XGBoost K=1)
- Pre-Processing (Reweighting)
- Post-Processing (ThresholdOpt)
- Hard Cluster-Predict v1 (K=2)
- Hierarchical Fair MoE v2 (K=2)

